

Statistics: Preliminary Ph.D. Course

Day 5: Sampling, Law of Large Numbers, and Central Limit Theorems

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Today's Session

We go back to Hansen's Book and then Simulate some nice results!

- **Independent and Identically Distributed (IID) Random Vectors, Random Sample, Population, and the DGP. Inference. Examples.**
- **Statistics, Parameters, and Estimators. The Analog/Plug-in Principle. Sampling Distribution and Standard Errors. The MSE: Bias and Variance. Examples.**
- **The Normal Sampling Model. Theorems and Proofs.**
- **Convergence in Probability. The Weak LLN. Consistency. The CMT. Almost Sure Convergence. The Strong LLN. Theorems, Proofs, and Examples. Simulations in R.**
- **Convergence in Distribution. Lévy's Continuity Theorem. Lindeberg-Lévy CLT. The CMT, Slutsky's Theorem, and the Delta Method. Theorems, Proofs, and Examples. Application: Asymptotic Distribution of Plug-in Estimators.**
- **Simulations in R.**

Sampling

Definitions

- **Statistics** is concerned with inference on parameters from data.
- **Definition 5.1:** The collection of random vectors $\{X_1, \dots, X_m\}$ are **Independent and Identically Distributed (IID)** if they are mutually independent with identical marginal distributions F .
- **Definition 5.2:** A **Data Set** is a collection of numbers, typically organized by **observation**, e.g., wage information on individuals. A data set is usually called a **Sample**.
- In statistical analysis we view the observations as realizations of random variables. This imposes structure and order on the data.
- **Definition 5.3:** A collection of random vectors $\{X_1, \dots, X_m\}$ is a **Random Sample** from the population F if X_i are independent and identically distributed with distribution F .
- **Definition 5.4:** The distribution F is called the population distribution or just the **Population**.

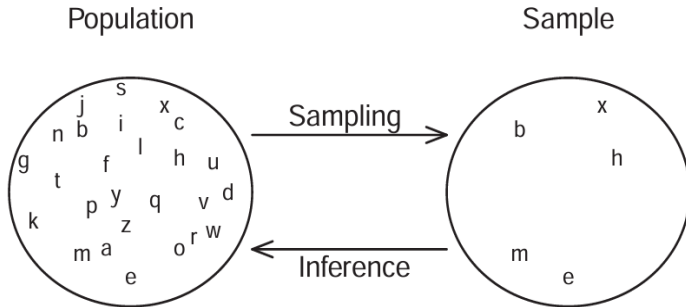
Sampling

Definitions

- **Definition 5.5:** A process by which an observation is created is called **Data Generating Process (DGP)**. In the DGP, the population is the probability model generating the data.
- **Definition 5.6:** The **Sample Size** n is the number of individuals in the sample, e.g., people, countries, time, etc.
- **Definition 5.7:** The **Inference Problem** is the problem of learning about the underlying process by examining the observations, i.e., based on a sample we want to infer properties of the originating population.
- The way in which the sample is collected affects the inference method to be used, as well as the properties of the parameters of interest. Random sampling provides a solid understanding, but there are other forms of sampling: time series data, panel data, etc.

Sampling

The Sampling Process: An Illustration



Sampling

Definitions

- The goal of statistics is to learn about population features, i.e., parameters.
- **Definition 5.8:** A **Parameter** θ is any function of the population F .
- **Definition 5.9:** A **Statistic** is a function of the sample $\{X_i\}_{i=1}^n$. Since it is a function of random variables, it is a random variable.
- Statistics are used to estimate parameters.
- An **Estimator** $\hat{\theta}$ for a parameter θ is a statistic intended as a guess about θ .
- We call $\hat{\theta}$ an **Estimate** when it is a realized value calculated in a specific sample.
- **Definition 5.10:** A standard way to construct an estimator is by the **Analog or Plug-in Principle**. The idea is to express the parameter θ as a function of the population F , and then express the estimator $\hat{\theta}$ as the analog function in the sample.

Sampling

Definitions

- **Definition 5.11:** The **Sample Mean** is the random variable (statistic) $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$.
- Many parameters can be written as the **expectation of a transformation** of X , i.e., $\theta = \mathbb{E}[g(X)]$ for some function g . The parameter is finite if g is integrable. The analog estimator of θ is just the sample mean:

$$\hat{\theta} = \frac{1}{n} \sum_{i=1}^n g(X_i)$$

- Other parameters can be written as **functions of population moments**, i.e., $\beta = h(\mathbb{E}[g(X)])$ for some functions g and h . The sample analog is

$$\hat{\beta} = h\left(\frac{1}{n} \sum_{i=1}^n g(X_i)\right)$$

Sampling

Definitions

- **Definition 5.11:** Statistics are random variables. Hence, statistics have probability distributions of their own. We call **Sampling Distribution** the distribution of a statistic because it is the distribution induced by sampling. It is unknown, in part because F is unknown.
- A considerable effort in statistical theory is to understand the sampling distribution of estimators. There are several approaches:
 1. Exact bias and variance.
 2. Exact distribution.
 3. Asymptotic approximations (as $n \rightarrow \infty$).
 4. Bootstrap.

We will focus on the first three. You will cover bootstrap in Econometrics I.

Sampling

Estimation Bias

- **Definition 5.12:** We say that an estimator is **Biased** if its sampling distribution is incorrectly centered (in terms of expectations). We define:

$$\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta$$

If the bias is zero, we say it is **Unbiased**.

Theorem 5.1 (Unbiasedness of the Sample Mean)

Let $\{X_i\}_{i=1}^n$ be IID. $\bar{X}_n$ is unbiased for $\theta = \mathbb{E}[X]$ if $\mathbb{E}[|X|] < \infty$.

Proof. (Hint: Use the definitions of expectation and sample mean).

- Even though unbiasedness seems a useful property for an estimator, it is one of many desirable properties. A “reasonable” and an “unbiased” estimator are not necessarily the same thing.

Example.

Sampling

Sampling Variance

- **Definition 5.13:** The variance is a key measure of the dispersion of a distribution. We often call the variance of an estimator, the **Sampling Variance**. It is typically associated with the precision of an estimator.

Theorem 5.2 (Sampling Variance of the Sample Mean)

Let $\{X_i\}_{i=1}^n$ be IID. $\mathbb{V}ar[\bar{X}_n] = \sigma^2/n$ if $\mathbb{E}[X^2] < \infty$, where $\sigma^2 = \mathbb{V}ar[X]$.

Proof. (Hint: Use the definitions of sample mean and IID).

- **Definition 5.14:** The Mean Squared Error (MSE) of an estimator $\hat{\theta}$ of θ is $MSE(\hat{\theta}) = \mathbb{E}[(\hat{\theta} - \theta)^2]$. It is a standard measure of the accuracy of an estimator.

Theorem 5.3 (Bias-Variance Decomposition)

For any estimator with finite variance: $MSE(\hat{\theta}) = \mathbb{V}ar[\hat{\theta}] + Bias^2(\hat{\theta})$.

Proof. (Hint: Use the definitions of MSE and Bias after adding and subtracting a term of interest).

Sampling

Estimation Bias

- **Definition 5.15:** The **Sample Variance** is the random variable (statistic)

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2.$$

Theorem 5.4 (Bias of Sample Variance)

Let $\{X_i\}_{i=1}^n$ be IID and $\mathbb{E}[X^2] < \infty$. The sample variance of X is biased downward, i.e., $\mathbb{E}[\hat{\sigma}^2] = (1 - \frac{1}{n})\sigma^2$

Proof. (Hint: Add and subtract $\mathbb{E}[X]$ and use the definitions of sample variance and IID).

- **Definition 5.16:** The **Re-scaled Sample Variance** is the random variable (statistic)

$$s^2 = \frac{n}{n-1} \hat{\sigma}^2.$$

Theorem 5.5 (Unbiasedness of Re-scaled Sample Variance)

Let $\{X_i\}_{i=1}^n$ be IID and $\mathbb{E}[X^2] < \infty$. The re-scaled sample variance of X is unbiased for $\sigma^2 = \text{Var}[X]$, i.e., $\mathbb{E}[s^2] = \sigma^2$.

Proof. (Hint: Use Theorem 5.4).

Sampling

Standard Errors

- **Definition 5.17:** A **Standard Error** $s.e(\hat{\theta})$ of an estimator $\hat{\theta}$ is the square root of an estimator $\hat{V}$, of $V = \mathbb{V}ar[\hat{\theta}]$.

Example. (Standard error for the sample mean).

- Standard errors are usually reported with parameter estimates to assess the precision of estimation. There is, in general, no reason to expect unbiasedness of standard errors.
- The previous discussion can be readily extended to the multivariate case by using vector-matrix notation.

Sampling

The Normal Sampling Model

- Notice that we have found the mean and variance of $\bar{X}_n$ under very general conditions. Without knowing its exact distribution. But we are also interested in its exact distribution.
- For any statistic, it is usually difficult to derive its exact distribution even if the exact distribution of the X_i 's were known.
- A traditional approach to deriving the exact sampling distribution of $\bar{X}_n$ is to assume that observations are normally distributed. This is convenient because an elegant exact theory can be derived.
- In practice, however, we rarely know the exact distribution of each observation.

Sampling

The Normal Sampling Model

Theorem 5.6 (Normal Sampling Model I)

Let $X_i \sim IIDN(\mu, \sigma^2)$. Then, $\bar{X}_n \sim N(\mu, \sigma^2/n)$.

Proof. (Hint: Use Theorem 4.3).

Theorem 5.7 (Normal Sampling Model II)

Let $\hat{e}_i = X_i - \bar{X}_n$. Then, $\mathbb{E}[\hat{e}_i] = 0$ and $\hat{e}_i \perp \bar{X}_n$.

Proof. (Hint: Use the definition of expectation and Theorem 4.2).

Theorem 5.8 (Normal Sampling Model III)

$\frac{(n-1)s^2}{\sigma^2} \sim \chi_{n-1}^2$ and it is independent of $\bar{X}_n$.

Proof. (See pg.145 in Hansen's Book for the first part, the second follows by independence and Theorem 5.7).

Sampling

The Normal Sampling Model

Theorem 5.9 (Normal Sampling Model IV: Gosset's Theorem)

The studentized ratio, t -statistic, or z -statistic, follows a Student- t distribution with $n - 1$ degrees of freedom:

$$T_n = \frac{\sqrt{n}(\bar{X}_n - \mu)}{s} \sim t_{n-1}$$

Proof. (Use Theorems 4.6 and 5.8).

- This is **one of the most famous results in statistical theory**, discovered by Gosset in 1908, at the time he was working at the Guinness Brewery!

Law of Large Numbers

Convergence in Probability

- What is the distribution of $\bar{X}_n$ when the observations are not normal? What is the distribution of estimators which are nonlinear functions of sample means? We can approximate it using **Asymptotic Theory**: LLNs and CLTs.
- We can approximate the sampling distributions of most econometric estimators.
- We first review the basics on convergence and limits of sequences.
- **Definition 5.18:** A sequence a_n has the **limit** a , written $a_n \rightarrow a$ as $n \rightarrow \infty$, or $\lim_{n \rightarrow \infty} a_n = a$, if for all $\delta > 0$ there is some $n_\delta < \infty$ such that for all $n \geq n_\delta$, $|a_n - a| \leq \delta$. In this case, we say that a_n converges to a as $n \rightarrow \infty$ (and the limit is unique).
- A sequence of numbers may converge, but what about a sequence of random variables? In what sense do they converge? *Examples.*

Law of Large Numbers

Convergence in Probability

- **Definition 5.19:** A sequence of random variables $Z_n \in \mathbb{R}$ **Converges in Probability** to c as $n \rightarrow \infty$, denoted $Z_n \xrightarrow{p} c$, or $\text{plim}_{n \rightarrow \infty} Z_n = c$, if for all $\delta > 0$,

$$\lim_{n \rightarrow \infty} \mathbb{P}(|Z_n - c| \leq \delta) = 1$$

We call c the **Probability Limit** or **plim** of Z_n . Alternatively, we can write:

$$\lim_{n \rightarrow \infty} \mathbb{P}(|Z_n - c| > \delta) = 0$$

Examples.

Law of Large Numbers

Weak Law of Large Numbers

Theorem 5.10 (The Weak Law of Large Numbers)

Let $X_i \in \mathbb{R}^m$ be IID and $\mathbb{E}[||X||] < \infty$, then as $n \rightarrow \infty$

$$\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{p} \mathbb{E}[X]$$

Proof.

- **Definition 5.20:** An estimator $\hat{\theta}$ of a parameter θ is **Consistent** whenever $\hat{\theta} \xrightarrow{p} \theta$ as $n \rightarrow \infty$.

Examples and Counterexamples.

Law of Large Numbers

The Continuous Mapping Theorem

Theorem 5.11 (The Continuous Mapping Theorem, CMT)

Let $Z_n \xrightarrow{p} c$ as $n \rightarrow \infty$, and let $h(\cdot)$ be a continuous function at point c . Then, $h(Z_n) \xrightarrow{p} h(c)$.

Examples.

- **Definition 5.21:** A sequence of random variables Z_n **Converges Almost Surely** to c as $n \rightarrow \infty$, denoted $Z_n \xrightarrow{a.s.} c$ whenever $\mathbb{P} \left(\lim_{n \rightarrow \infty} Z_n = c \right) = 1$.
- It is stronger than convergence in probability in the sense that *a.s.* convergence implies convergence in probability.

Theorem 5.12 (The Strong Law of Large Numbers)

Let $X_i \in \mathbb{R}^m$ be IID and $\mathbb{E}[||X||] < \infty$, then as $n \rightarrow \infty$, $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i \xrightarrow{a.s.} \mathbb{E}[X]$.

Central Limit Theorem

Convergence in Distribution

- **Definition 5.22:** Let Z_n be a sequence of random variables or vectors with distribution $G_n(u) = \mathbb{P}(Z_n \leq u)$. We say that Z_n **Converges in Distribution** to Z as $n \rightarrow \infty$, denoted $Z_n \xrightarrow{d} Z$, if for all u at which $G(u) = \mathbb{P}(Z \leq u)$ is continuous, $G_n(u) \rightarrow G(u)$ as $n \rightarrow \infty$.
- We also say that G_n converges weakly to G . It is common to refer to Z and its distribution $G(u)$ as the **Asymptotic Distribution**, large sample distribution or limit distribution of Z_n .
- When the limit distribution is degenerate, i.e., $\mathbb{P}(Z = c) = 1$ for some c , we can write the convergence as $Z_n \xrightarrow{d} c$ which is **equivalent** to $Z_n \xrightarrow{p} c$.

Motivation with the Sample Mean.

Theorem 5.13 (Lévy's Continuity Theorem)

$Z_n \xrightarrow{d} Z$ if $\mathbb{E}[\exp(itZ_n)] \rightarrow \mathbb{E}[\exp(itZ)]$ for every $t \in \mathbb{R}$ as $n \rightarrow \infty$, for IID random variables Z_n .

Central Limit Theorem

Classic CLT

Theorem 5.14 (Lindeberg-Lévy Central Limit Theorem, CLT)

Let X_i be IID and $\mathbb{E}[X^2] < \infty$, then as $n \rightarrow \infty$,

$$\sqrt{n} \left(\frac{\bar{X}_n - \mu}{\sigma} \right) \xrightarrow{d} N(0, 1)$$

where $\mu = \mathbb{E}[X]$ and $\sigma^2 = \mathbb{E}[(X - \mu)^2]$.

Proof. (Hint: Use Theorem 5.13 and the Taylor Series).

Central Limit Theorem

Multivariate CLT

Examples.

- The CLT says that the distribution of the random variable $\sqrt{n}(\bar{X}_n - \mu)$ is approximately the same as $N(0, \sigma^2)$ when n is large. Sometimes we write as an **informal interpretative statement**: $\bar{X}_n \underset{a}{\sim} N(\mu, \sigma^2/n)$.
- The classic CLT has a **multivariate generalization**.

Theorem 5.15 (Multivariate Lindeberg-Lévy Central Limit Theorem, CLT)

Let $X_i \in \mathbb{R}^m$ be IID and $\mathbb{E}[||X||^2] < \infty$, then as $n \rightarrow \infty$,

$$\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{d} N(0, \Sigma)$$

where $\mu = \mathbb{E}[X]$ and $\Sigma = \mathbb{E}[(X - \mu)(X - \mu)']$.

Central Limit Theorem

CMT, Slutsky and DM

Theorem 5.16 (Continuous Mapping Theorem II, CMT)

Let Z_n be IID. If $Z_n \xrightarrow{d} Z$ as $n \rightarrow \infty$, and $h : \mathbb{R}^m \rightarrow \mathbb{R}^k$ has the set of discontinuity points D_h such that $\mathbb{P}(Z \in D_h) = 0$, then $h(Z_n) \xrightarrow{d} h(Z)$ as $n \rightarrow \infty$.

- A special case is Slutsky's Theorem (a.k.a. Lemma):

Theorem 5.17 (Slutsky's Theorem)

Let Z_n be IID. If $Z_n \xrightarrow{d} Z$ and $c_n \xrightarrow{p} c$ as $n \rightarrow \infty$, then

1. $Z_n + c_n \xrightarrow{d} Z + c$
2. $Z_n c_n \xrightarrow{d} Zc$
3. $\frac{Z_n}{c_n} \xrightarrow{d} \frac{Z}{c}$ if $c \neq 0$.

Central Limit Theorem

CMT, Slutsky and DM

- We can now state the final and **one of the most important results** in asymptotic theory. It is a direct consequence of the Mean Value Theorem from calculus.

Theorem 5.18 (The Delta Method, DM)

If $\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{d} \xi$ as $n \rightarrow \infty$ and $h(u)$ is continuously differentiable in a neighborhood of θ , then as $n \rightarrow \infty$

$$\sqrt{n}(h(\hat{\theta}) - h(\theta)) \xrightarrow{d} H(\theta)\xi$$

where $H(u) = \frac{\partial}{\partial u} h(u)$ is the Jacobian of h . In particular, if $\xi \sim N(0, V)$ then as $n \rightarrow \infty$

$$\sqrt{n}(h(\hat{\theta}) - h(\theta)) \xrightarrow{d} N(0, H(\theta) V H(\theta)')$$

Proof. (Hint: Use the Mean Value Theorem).

Examples. Application: Asymptotic Distribution of Plug-in Estimators.

LLN and CLT

Simulations in R

- We now **simulate the LLN and the CLT** to fully grasp their power.
- I will provide some **R code** in case you want to replicate it at home.
- We will need the following libraries for the plotting, so you may need to install them.

```
library(ggplot2)  
library(ggthemes)  
library(gridExtra)
```

- I now provide some **code** for the LLN.

Law of Large Numbers

Simulations in R

```
set.seed(547)
# Create a vector of increasing sample sizes.

nsize <- seq(10, 100000, 100)

# Data from a Normal(0,100).

average_x1 <- c(1:length(nsize))      # Create a vector of averages for each n.
pop_mean <- rep(0, length(nsize))     # Fix the true population mean for the graph.

for(i in 1:1000){
  x1 <- rnorm(nsize[i], 0, 100)       # Generate the data from the population.
  average_x1[i] <- mean(x1)
}

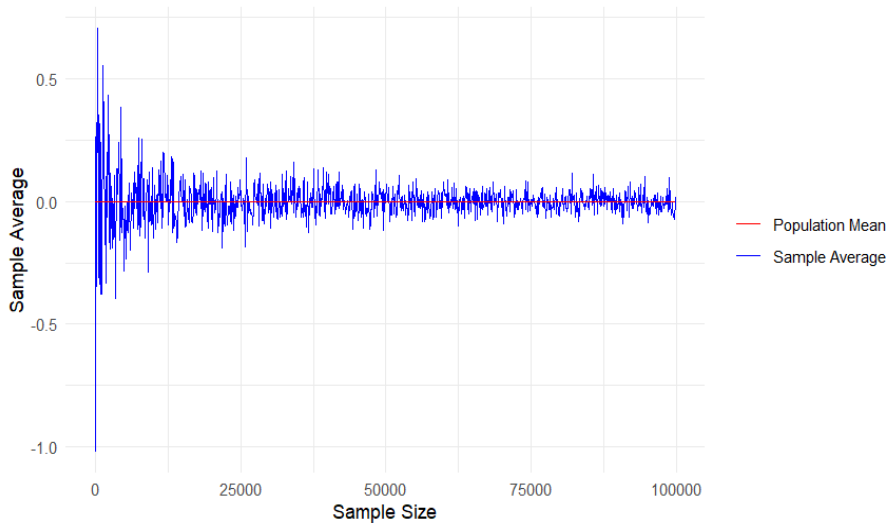
# Plot it to see the convergence.

lln_1 <- as.data.frame(cbind(nsize, average_x1, pop_mean))
colnames(lln_1) <- c("nsize", "average", "truemean")
plot1_lln <- ggplot(lln_1, aes(x = nsize)) + geom_line(aes(y = average, color = "Sample Average")) + geom_line(aes(y = truemean, color = "Population Mean")) + labs(title = "Law of Large Numbers: Simulation 1", x = "Sample Size", y = "Sample Average") + scale_color_manual(values = c("Sample Average" = "blue", "Population Mean" = "red"), guide = guide_legend(title = NULL)) + theme_minimal()
```

Law of Large Numbers

Simulations in R

Law of Large Numbers: Simulation 1



Law of Large Numbers

Simulations in R

```
# Data from a Poisson(5).

average_x2 <- c(1:length(nsize))      # Create a vector of averages for each n.
pop_mean2 <- rep(5, 1000)             # Fix the true population mean for the graph.

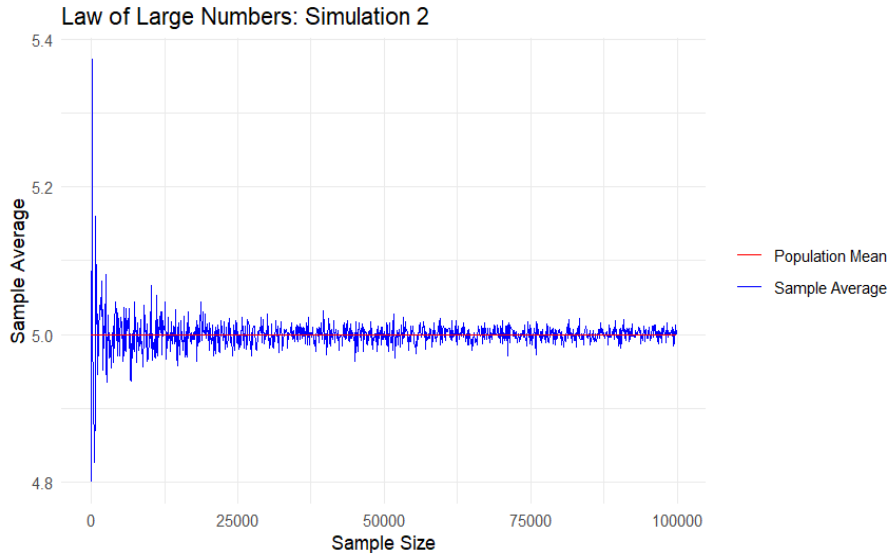
for(i in 1:1000){
  x2 <- rpois(nsize[i], 5)
  average_x2[i] <- mean(x2)
}

# Plot it to see the convergence.

lln_2 <- as.data.frame(cbind(nsize, average_x2, pop_mean2))
colnames(lln_2) <- c("nsize", "average", "truemean")
plot2_lln <- ggplot(lln_2, aes(x = nsize)) + geom_line(aes(y = average, color = "Sample Average")) + geom_line(aes(y = truemean, color = "Population Mean")) + labs(title = "Law of Large Numbers: Simulation 2", x = "Sample Size", y = "Sample Average") + scale_color_manual(values = c("Sample Average" = "blue", "Population Mean" = "red"), guide = guide_legend(title = NULL)) + theme_minimal()
```

Law of Large Numbers

Simulations in R



Law of Large Numbers

Simulations in R

```
# Data from a Cauchy(5,10).

average_x3 <- c(1:length(nsize))      # Create a vector of averages for each n.
                                       # Recall that there is no population mean.

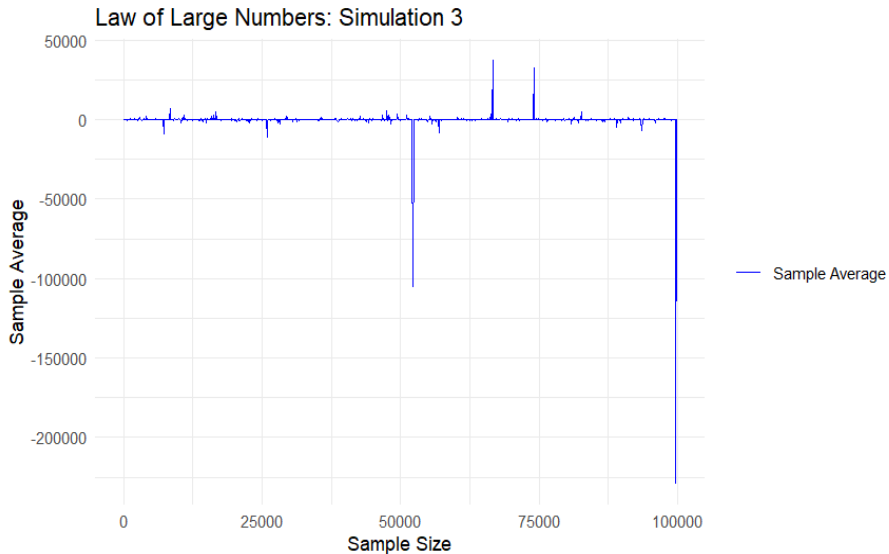
for(i in 1:1000){
  x3 <- rcauchy(nsize[i], 5, 100)
  average_x3[i] <- mean(x3)
}

# Plot it to see... nothing.

lln_3 <- as.data.frame(cbind(nsize, average_x3))
colnames(lln_3) <- c("nsize", "average")
plot3_lln <- ggplot(lln_3, aes(x = nsize)) + geom_line(aes(y = average, color = "Sample Average")) + labs(title = "Law of Large Numbers: Simulation 3", x = "Sample Size", y = "Sample Average") + scale_color_manual(values = c("Sample Average" = "blue"), guide = guide_legend(title = NULL)) + theme_minimal()
```

Law of Large Numbers

Simulations in R



Law of Large Numbers

Simulations in R

```
# Data from a Uniform[0,1].

average_x4 <- c(1:length(nsize))      # Create a vector of averages for each n.
pop_mean3 <- rep(0.5, 1000)           # Fix the true population mean for the graph.

for(i in 1:1000){
  x4 <- runif(nsize[i], 0, 1)
  average_x4[i] <- mean(x4)
}

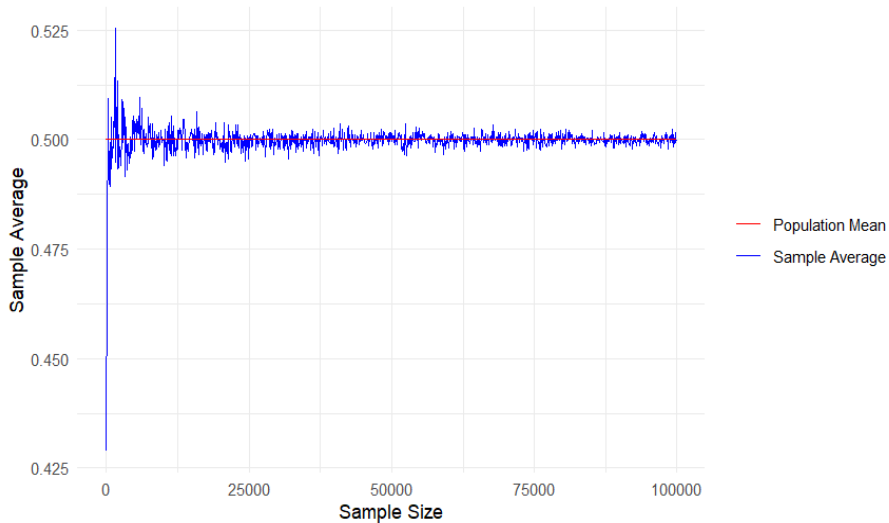
# Plot it to see the convergence.

lln_4 <- as.data.frame(cbind(nsize, average_x4, pop_mean3))
colnames(lln_4) <- c("nsize", "average", "truemean")
plot4_lln <- ggplot(lln_4, aes(x = nsize)) + geom_line(aes(y = average, color = "Sample Average")) + geom_line(aes(y = truemean, color = "Population Mean")) + labs(title = "Law of Large Numbers: Simulation 4", x = "Sample Size", y = "Sample Average") + scale_color_manual(values = c("Sample Average" = "blue", "Population Mean" = "red"), guide = guide_legend(title = NULL)) + theme_minimal()
```

Law of Large Numbers

Simulations in R

Law of Large Numbers: Simulation 4



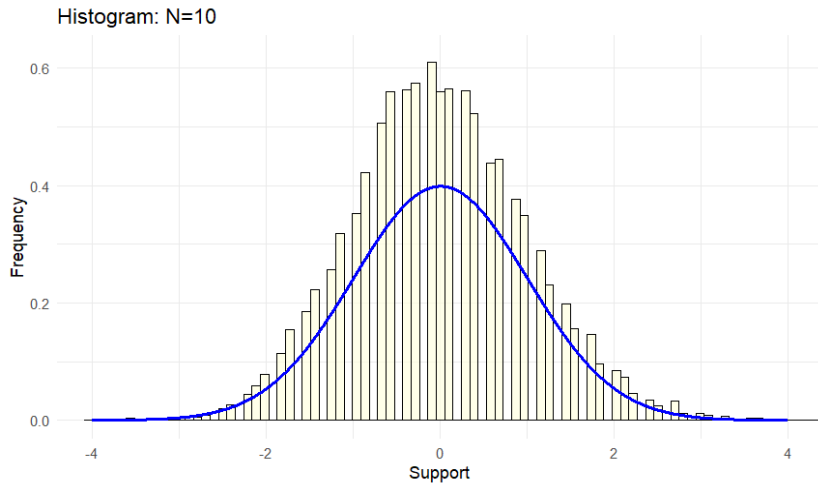
Central Limit Theorem

Simulations in R

- We simulate the CLT using data from a Poisson distribution with $\lambda = 5$.
- Let's see the CLT in action!

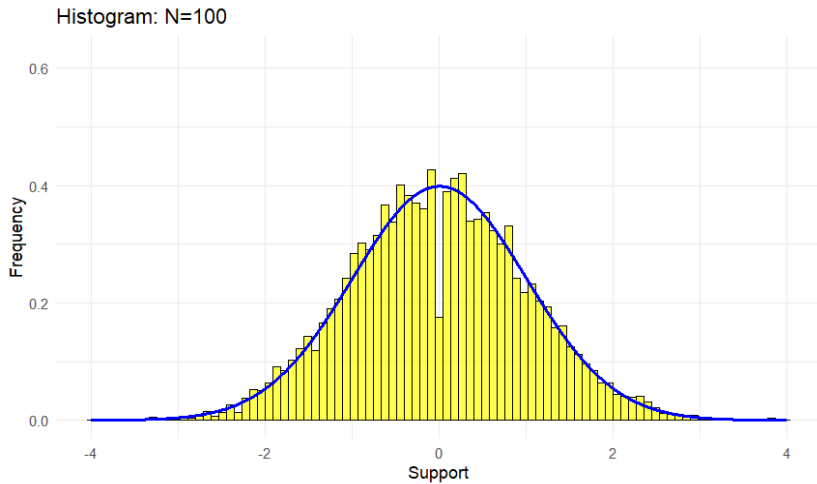
Central Limit Theorem

Simulations in R



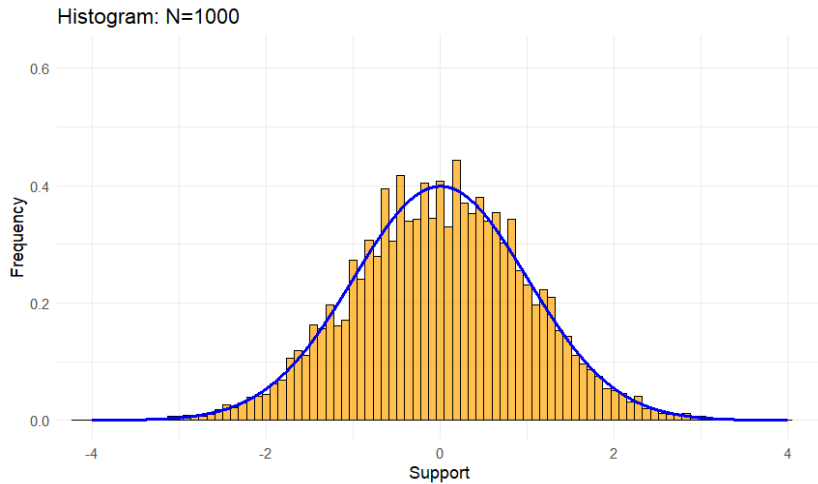
Central Limit Theorem

Simulations in R



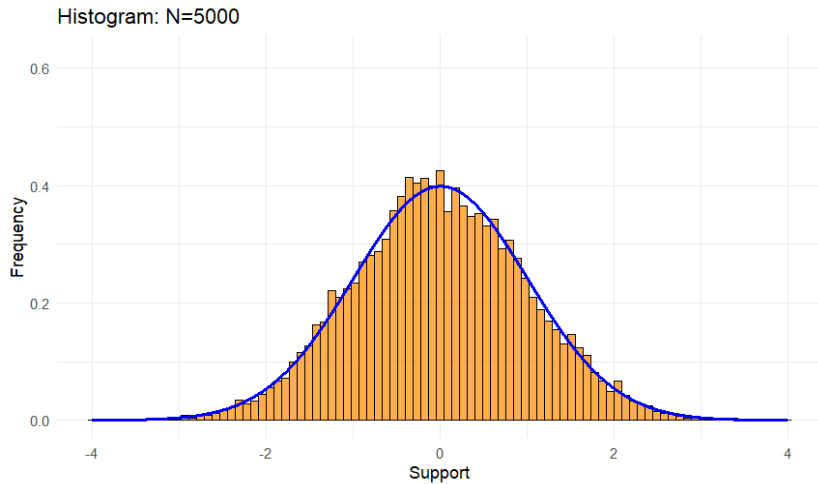
Central Limit Theorem

Simulations in R



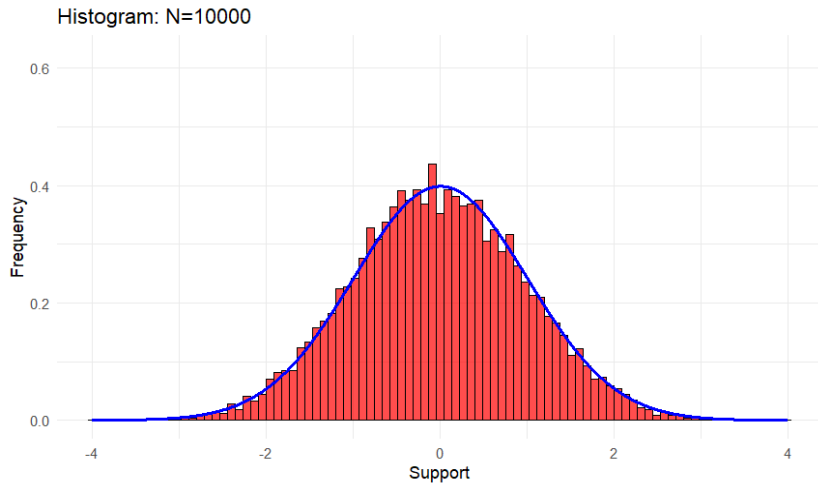
Central Limit Theorem

Simulations in R



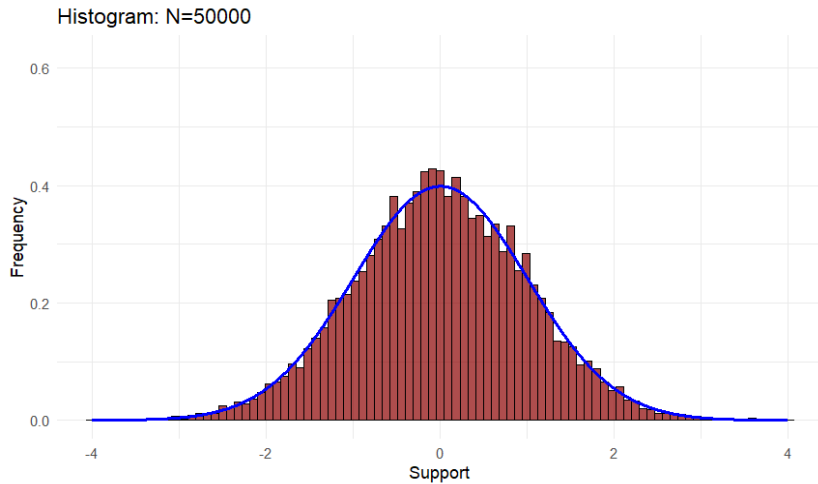
Central Limit Theorem

Simulations in R



Central Limit Theorem

Simulations in R



Recommended Exercises

Hansen's Book

- Chapter 6, Exercises: 6.1-6.10 and 6.12-6.15.
- Chapter 7, Exercises: 7.1-7.4, 7.5a-b, 7.5h, 7.5j-k, and 7.6-7.11.
- Chapter 8, Exercises: 8.1-8.4 and 8.6-8.10.

References

Hansen, B. (2022). Probability and Statistics for Economists. *Princeton University Press*. Chapters 6, 7, and 8.

Hayashi, F. (2000). Econometrics. *Princeton University Press*. Chapter 2.

Wasserman, L. (2004). All of Statistics: A Concise Course in Statistical Inference. *Springer*.

Shao, J. (1999). Mathematical Statistics. *Springer*.